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Synthesized White Paper: Exploring UAV-Based Cloud Seeding as a Preventative Strategy Against Wildfires in Fort McMurray

Executive Summary

This white paper examines a proactive wildfire prevention strategy for Fort McMurray involving cloud seeding technology—an atmospheric process that enhances precipitation—deployed via unpiloted aerial vehicles (UAVs). In response to the increasing frequency and severity of wildfires in the region [1], the proposed approach evaluates the feasibility and broader implications of UAV-based cloud seeding before and during wildfire season [2].

The analysis is structured around four key dimensions: disaster resilience, health and well-being, nature and biodiversity, and infrastructure [3]. From a disaster resilience perspective, UAV-based cloud seeding can reduce wildfire risk by increasing localized rainfall, decreasing ignition potential, and limiting fire spread [4]. The strategy also supports the development of accountable governance systems and strengthens multi-level coordination involving government, civil society, and private actors [2].

In terms of health and well-being, the approach helps reduce exposure to wildfire smoke [5] and extreme heat [6], while addressing secondary health effects such as food system disruption and mental stress [7]. This reduces strain on healthcare services and improves overall community resilience [3].

Ecological assessments suggest that, when managed appropriately, cloud seeding poses minimal environmental risk [8, 9]. It also contributes to adaptation efforts by helping ecosystems recover from fire-related damage and maintain long-term viability [10].

From an infrastructure standpoint, the approach aligns with climate-responsive planning principles and supports “all stages of the infrastructure lifecycle from design to maintenance” [11]. While the effectiveness of cloud seeding depends on favorable atmospheric conditions [12] and requires ongoing field validation, it represents a scalable and environmentally sound option for wildfire mitigation.

This white paper synthesizes recent scientific research, regional case studies, and technical evaluations to demonstrate how UAV-based cloud seeding can enhance wildfire prevention efforts, protect public health, conserve ecosystems, and strengthen climate-resilient infrastructure in alignment with Canada’s National Adaptation Strategy [3].

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Introduction and Background

The Fort McMurray region in northeastern Alberta has become increasingly vulnerable to large-scale wildfires, primarily due to shifting climate conditions [13]. The 2016 Fort McMurray wildfire—referred to as “The Beast” [14]—led to the evacuation of nearly 90,000 residents and the destruction of approximately 2,400 structures, making it one of the most expensive natural disasters in Canadian history [15]. Since then, wildfire frequency and intensity in the region have continued to increase [1], placing growing pressure on emergency response systems and threatening lives, public health, infrastructure, and ecosystems [2]. As fire seasons become more unpredictable, there is a pressing need for proactive, scalable wildfire prevention strategies.

This white paper examines the potential of UAVs equipped with cloud seeding technology as a proactive wildfire mitigation strategy. Cloud seeding involves dispersing particles such as silver iodide or potassium chloride into cloud systems to induce rainfall [12], lowering temperatures and increasing soil moisture to reduce ignition risk. When deployed via UAVs, this technique enables precise and efficient application, particularly in remote or high-risk areas, and complements existing fire prevention methods [2].

The approach aligns with the goals of Canada’s National Adaptation Strategy [3] by supporting proactive risk reduction, enhancing community well-being, promoting environmental stewardship, and informing climate-resilient infrastructure planning. These priorities are explored through four interconnected systems:

- **Disaster resilience:** Evaluates how UAV-based cloud seeding can support early intervention, improve coordination, and strengthen governance mechanisms for wildfire risk reduction, involving cooperation across emergency management, local authorities, and regional planning networks [13].
- **Health and well-being:** Considers the reduction of health risks related to wildfire smoke, heat exposure, and mental stress, with implications for public health planning, service delivery, and emergency preparedness across communities [7].
- **Nature and biodiversity:** Explores how limiting fire spread can protect natural systems and support ongoing efforts in ecosystem monitoring, restoration, and land management [14].
- **Infrastructure:** Assesses the integration of cloud seeding into infrastructure systems designed to withstand climate risks, informing planning, design, and maintenance through climate-informed technical standards [15].

By considering these interconnected systems, this paper demonstrates that UAV-based cloud seeding is not only technically feasible but also supports broader resilience goals. It offers a scalable, adaptive strategy to protect communities, ecosystems, and infrastructure in Fort McMurray and other wildfire-prone regions.

Solution Analysis Through Various Objectives

To evaluate the feasibility and broader impact of UAV-based cloud seeding, the following section examines its role across four interconnected systems: disaster resilience, health and well-being, nature and biodiversity, and infrastructure.

A. Disaster Resilience

Improving wildfire resilience in high-risk regions such as Fort McMurray requires the development of governance systems that are efficient, transparent, and capable of coordinated action [3]. Incorporating UAV-based cloud seeding into wildfire mitigation strategies can enhance early-stage risk reduction, strengthen inter-agency collaboration, and support broader community engagement [2].

Although UAV-based cloud seeding is not yet widely adopted for wildfire prevention, its potential as a proactive moisture-enhancement strategy could reduce ignition risk under increasingly unstable weather conditions [16]. The technology is particularly valuable in remote or underserved areas, where traditional firefighting resources may be limited [2]. When implemented strategically, cloud seeding can reduce vegetation dryness, delay fire ignition, and provide critical lead time for mobilizing response efforts [12].

UAV-based cloud seeding supports wildfire resilience by helping reduce environmental risks and working within existing policies and governance structures [13]. Table 1 shows the key environmental and policy factors that affect wildfire resilience, highlighting the need to connect new technologies with established systems for planning and decision-making [17].

Table 1. Factors affecting wildfire risk and resilience in high-latitude communities [17]

Factor	Description
Environmental conditions	Climate patterns influencing wildfire frequency and intensity
Community preparedness	Local initiatives and infrastructure to prevent and respond to wildfires
Land use practices	Management strategies affecting vegetation and fuel loads
Socioeconomic status	Economic resources impacting recovery and mitigation efforts
Policy and governance	Regulations and policies guiding wildfire management and response

To align with the whole-of-society approach in Canada’s National Adaptation Strategy [3], deploying UAV-based cloud seeding requires collaboration between governments, researchers, and local communities. Its successful use in international drought mitigation [7] shows potential for use in wildfire prevention, though further field testing is needed to understand its long-term effectiveness.

B. Health and Well-Being

Beyond reducing wildfire risk at the system level, UAV-based cloud seeding also offers direct benefits to human health. Canada's National Adaptation Strategy underscores the need to protect communities from climate-related health risks, including wildfire smoke, extreme heat, and mental health stressors [3]. In this context, UAV-based cloud seeding offers a promising approach to reducing public health impacts from increasingly frequent and severe wildfires.

The growing intensity of wildfires in Canada presents a serious public health challenge [18, 19]. Smoke exposure contributes to respiratory illnesses such as asthma and chronic bronchitis, along with increased emergency visits and, in severe cases, premature mortality among vulnerable populations [5, 6]. Wildfires also compromise water quality by introducing ash and debris into supply systems [20], while evacuations can cause elevated stress, food insecurity, and disruption of care networks [21]. Figure 1 illustrates residents filtering smoke from a wildfire, emphasizing the need for effective mitigation [22].



Fig. 1. Two ladies trying to cover their mouth and nose to filter out smoke from a wildfire [21]

UAV-based cloud seeding can help address these risks by lowering wildfire ignition potential. Eco-safe agents like sodium iodide increase cloud condensation nuclei, enhancing rainfall and soil moisture [16, 23]. This reduces vegetation flammability and supports both fire prevention and post-fire recovery [24]. Precipitation also assists in atmospheric cleansing, with larger raindrops removing harmful particulates from the air [25], supporting respiratory health and environmental resilience [26].

Beyond prevention, UAVs equipped with thermal sensors and real-time imaging improve wildfire monitoring and early response [27]. Early detection shortens fire duration and limits secondary health impacts such as extreme heat exposure, air pollution, and mental stress [26]. Real-time data also enhances coordination between emergency services and healthcare teams, strengthening situational awareness and response capacity [28].

Integrating UAV-based wildfire monitoring with public health preparedness offers a scalable strategy for early intervention and long-term resilience. As wildfire seasons continue to intensify, such systems provide a proactive means to safeguard health and reduce pressure on emergency and healthcare infrastructure.

C. Nature and Biodiversity

In addition to supporting public health, cloud seeding contributes to environmental protection goals. Protecting Canada's ecosystems and biodiversity is another key component of the National Adaptation Strategy [3]. To support this goal, cloud seeding operations must be environmentally responsible and adapted to local conditions.

One of the main environmental concerns in cloud seeding is the application of silver iodide, which has the potential to accumulate in natural systems [12]. However, there are researches indicate that the levels of silver iodide used in typical operations are very low and remain well under recognized toxicity thresholds [29, 30]. Table 2 outlines findings from an environmental study on this issue [29].

Table 2. Silver iodide (AgI) concentrations from cloud seeding and environmental impact [29]

Parameter	Typical AgI concentration in precipitation (µg/L)	Natural background level (µg/L)	Toxicity threshold for aquatic life (µg/L)	Implication for nature/biodiversity
Unseeded rainfall	0.0	0.01 - 0.1	100	Baseline condition, no impact
Seeded rainfall (low dose)	0.05 - 0.1	0.01 - 0.1	100	Negligible increase, well below toxic levels
Seeded rainfall (high dose)	0.1 - 0.5	0.01 - 0.1	100	Slightly elevated, still safe for ecosystems
Soil accumulation (annual)	0.001 - 0.01 (µg/g)	0.005 - 0.02 (µg/g)	50 (µg/g)	No significant bioaccumulation

While the use of silver iodide within operational guidelines appears safe, other ecological factors must be considered. Excessive seeding could oversaturate soils, affecting root structures and the dynamics of local vegetation [31]. Changes in soil moisture may also alter habitat conditions, influencing species diversity [8]. To manage these risks, seeding protocols should incorporate real-time environmental monitoring, precise precipitation targeting, and timing restrictions during sensitive periods such as breeding or migration.

Cloud seeding also alters cloud formation processes and the timing of rainfall, with effects that can persist for several days [32]. These changes may interfere with seasonal precipitation patterns, especially for species adapted to short, seasonal precipitation periods [9]. To prevent stress on vulnerable ecosystems, seeding should be avoided in areas that are ecologically fragile or already under hydrological strain.

When applied with environmental safeguards and adaptive management, UAV-based cloud seeding can help reduce the ecological impacts of wildfires while supporting species adaptation and ecosystem resilience.

D. Infrastructure

Following its environmental applications, UAV-based cloud seeding also contributes to building climate-resilient infrastructure systems. Canada's climate adaptation framework prioritizes infrastructure risk assessment and adaptation strategies such as emergency planning, fire-resistant construction, and vegetation control [33]. However, current measures often do not properly address drought, a significant factor in the possibility of wildfire happening [16]. In areas that are prone to drought, UAV-based cloud seeding presents an effective solution by enhancing rainfall [15].

Compared to traditional cloud seeding methods that rely on expensive and high-risk crewed aircraft, UAV-based systems offer a safer, more cost-effective, and more precise alternative [33]. As shown in Figure 2, UAV platforms are much simpler in design and operation than research aircraft, which significantly reduces both logistical complexity and overall cost. While aircraft require specialized crews and equipment for safe operation, UAVs can be deployed with minimal personnel and infrastructure, making them a practical option for remote or high-risk environments [16].

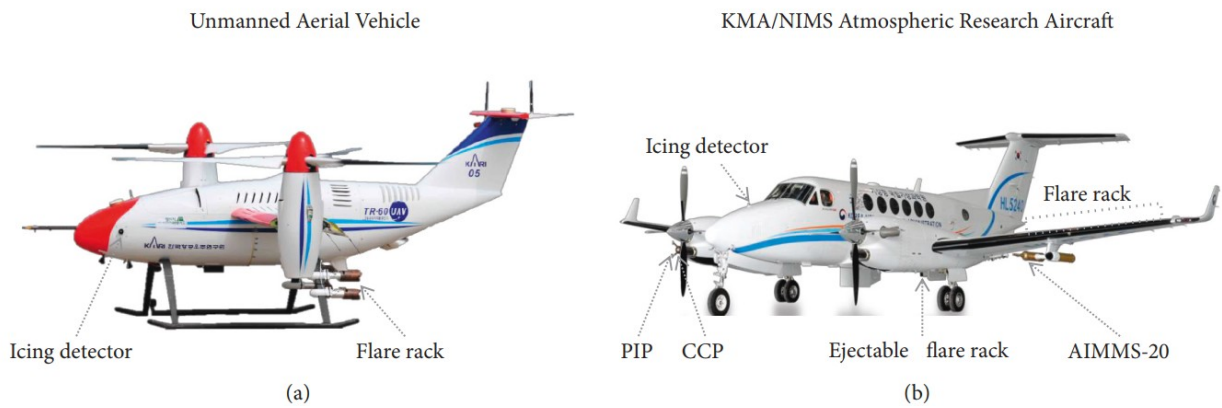


Fig. 2. Comparison of a UAV and an aircraft [16]

In addition to preventing wildfires, UAV-based cloud seeding also helps reduce wildfire impact by combining automation, atmospheric data collection, and targeted deployment [16]. Integrating this technology into Canada's broader infrastructure adaptation plans enables earlier intervention before fire conditions escalate, enhancing the overall effectiveness of wildfire response strategies [34].

The approach covers all stages of infrastructure planning and management, including site selection, design, adaptation, emergency response, and long-term maintenance [34]. Its implementation requires cost-benefit analysis, environmental assessment, and additional field testing [29]. UAV-based operations can be adapted to high-risk areas and integrated into local and regional emergency systems, helping to build more climate-resilient infrastructure [16].

Overall, UAV-based cloud seeding provides a practical and flexible way to prevent wildfires and make infrastructure more resilient to climate-related risks.

Conclusion

As climate change amplifies wildfire risk across northern wildfire-prone regions, Fort McMurray remains increasingly vulnerable to recurring, high-impact fire events [18]. This report has assessed the potential of UAV-based cloud seeding as an emerging tool for wildfire prevention, with a focus on its ability to lower ignition risk by increasing pre-season rainfall. While still in early stages of implementation, UAV-based cloud seeding offers a proactive complement to traditional suppression methods, helping move from reacting to preventing fires [16]. UAV-based cloud seeding aligns with national adaptation priorities by contributing to risk reduction, public health protection, ecosystem resilience, and climate-informed infrastructure planning [3].

From a disaster resilience perspective, this technology can fit within existing governance systems and improve coordination among public agencies, private organizations, and local communities [13]. This collaborative model helps build local capacity and reduces wildfire impacts, especially for vulnerable or high-risk populations [17].

In terms of health and well-being, limiting wildfire occurrence helps mitigate secondary effects such as smoke exposure, respiratory illness, mental health strain, and disruption of traditional food systems [19]. A prevention-based approach helps strengthen public health systems over time as climate conditions continue to change.

For nature and biodiversity, UAV-based cloud seeding is a low-impact method that helps protect sensitive ecosystems from large wildfires [8]. By reducing damage, it supports ecosystem monitoring, recovery, and adaptive management, helping maintain long-term ecological stability [29].

Finally, the approach fits with modern infrastructure standards that include climate adaptation in planning, design, and operations. UAVs can be used within regional emergency systems and focused on high-risk areas, helping make infrastructure more resilient throughout its entire lifecycle [34].

Although the use of UAV-based cloud seeding for wildfire prevention is still in early stages, current evidence shows it has strong potential as part of a broader resilience strategy. Ongoing research, field testing, and coordinated planning will be key to scaling the technology effectively. As climate risks continue to grow, investing in innovative, data-driven tools such as UAV-based cloud seeding will be important for protecting people, ecosystems, and infrastructure in Fort McMurray and other vulnerable areas.

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